

Functions & Procedures (Algo + C)

Statements with solutions in pseudo-algorithm and C language

Exercise 1 — Linear function

Statement. Write a function $f(x) = 3x + 2$ that returns the value for an integer x .

Exercise 2 — Swap procedure

Statement. Write a *procedure* that swaps two integers.

Exercise 3 — Even predicate

Statement. Write a boolean function `is_even(n)`.

Exercise 4 — Maximum of three (function)

Statement. Write `max3(a,b,c)`.

Exercise 5 — Factorial (recursive)

Statement. Write a recursive `fact(n)` ($n \geq 0$).

Exercise 6 — Fibonacci (recursive)

Statement. Write `fib(n)` which returns the n -th term (0,1,1,2,3,...).

Exercise 7 — Fast exponentiation (binary exponentiation)

Statement. Write `pow_int(a,n)` for integers a , $n \geq 0$.

Exercise 8 — GCD (recursive Euclid)

Statement. Write a recursive `gcd(a,b)`.

Exercise 9 — LCM using GCD

Statement. Write `lcm(a,b)` using `gcd`.

Exercise 10 — Sum and average of an array

Statement. Functions `sum(T,n)` and `average(T,n)`.

Exercise 11 — Min & Max with output parameters (procedure)

Statement. Procedure `minmax(T,n, min, max)`.

Exercise 12 — Sequential search

Statement. `index_of(T,n,x)` returns the index or `-1`.

Exercise 13 — Binary search (sorted array)

Statement. `bsearch_int(T,n,x)` returns the index or `-1`.

Exercise 14 — Bubble sort (procedure)

Statement. Procedure `bubble_sort(T,n)`.

Exercise 15 — Vowel count (string function)

Statement. `nb_vowels(s)` ignoring case.

Exercise 16 — Palindrome (function)

Statement. `is_palindrome(s)` (ignore spaces and case).

Exercise 17 — String reversal (procedure)

Statement. `reverse(s)` reverses a string in place.

Exercise 18 — Matrix: trace (function) and transposition (procedure)

Statement. For a square $N \times N$ matrix, write `trace(A,N)` and `transpose(A,N)`.

Exercise 19 — Variance (function)

Statement. Write `variance(T,n)` (population variance) and `std_dev`.

Exercise 20 — Primality test & next prime

Statement. Write `is_prime(n)` and `next_prime(n)`.